

DEPARTMENT OF ELECTRICAL ENGINEERING

Electronic Devices & Circuits Laboratory (EDC LAB)

EXPERIMENT 4: NMOS Common-Source Amplifier with Resistive Load: DC Transfer Characteristic and Small-Signal Gain

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1. Aim and Objectives

1. To construct an N-channel enhancement MOSFET Common-Source (CS) amplifier with a resistive drain load (R_D).
2. To measure, record, and plot its full **DC Transfer Characteristic** (V_{OUT} vs V_{IN}) by sweeping V_{IN} from 0 V to 2.4 V under a supply voltage $V_{DD} = 10$ V, identifying the cutoff, saturation (amplification), and triode regions.
3. To extract the small-signal voltage gain $A_v(V_{IN})$ as a function of the DC bias point using two independent methods:

- **Method 1 (Numerical Slope):** Differentiating the measured DC transfer curve:

$$A_v(V_{IN}) = \frac{dV_{OUT}}{dV_{IN}}$$

- **Method 2 (Direct AC Measurement):** Superposing a sinusoidal test tone ($v_{in} = 10$ mV amplitude, $f = 1$ kHz) through a $C = 0.1$ μ F coupling capacitor, sweeping the DC bias offset via a 1 M Ω isolation resistor from 0 V to 1.8 V, and measuring the AC output response ($V_{OUT(RMS)}$).

2. Apparatus and Software Tools Required

Table 1: Equipment, Components, and Simulation Tools Used

S.No.	Component / Equipment	Specification / Value	Quantity
1	N-Channel MOSFET	2N7002 / 2N7000/N7000	1
2	Drain Load Resistor (R_D)	10 k Ω (Part A) / 300 k Ω (Part B)	1 each
3	DC Bias Isolation Resistor	1 M Ω (Part B)	1
4	AC Coupling Capacitor	0.1 μ F ceramic/film	1
5	Regulated Dual DC Power Supply	0–30 V (for $V_{DD} = 10$ V and variable DC bias)	1
6	Function / Signal Generator	Sine Wave, 10 mV peak (20 mV _{pp}), 1 kHz	1
7	Digital Oscilloscope / DMM	AC RMS & peak-to-peak voltage measurement	1
8	Simulation Software	LTspice XVII	–

3. Overall Theoretical Framework

In a Common-Source (CS) NMOS amplifier with a resistive drain load R_D , the input voltage V_{IN} is applied between the gate and source terminals ($V_{GS} = V_{IN}$), while the output voltage V_{OUT} is taken at the drain node ($V_{DS} = V_{OUT} = V_{DD} - I_D R_D$).

Depending on the magnitude of V_{IN} relative to the device threshold voltage (V_T) and the drain voltage V_{OUT} , the transistor traverses three distinct operational regions:

1. **Cut-off Region** ($V_{IN} < V_T$):

$$I_D = 0 \implies V_{OUT} = V_{DD} - 0 \cdot R_D = V_{DD} \quad (1)$$

The output remains pinned at the positive rail ($V_{DD} = 10$ V). The incremental transconductance is zero ($g_m = 0$), yielding zero small-signal gain:

$$A_v = \frac{dV_{OUT}}{dV_{IN}} = 0 \quad (2)$$

2. **Saturation Region** ($V_{IN} \geq V_T$ and $V_{OUT} \geq V_{IN} - V_T$): The inversion channel pinches off before reaching the drain terminal. The drain current obeys the classic square-law relation:

$$I_D = \frac{1}{2} k_n (V_{IN} - V_T)^2 (1 + \lambda V_{OUT}) \approx \frac{1}{2} k_n (V_{IN} - V_T)^2 \quad (3)$$

The corresponding DC output voltage drops quadratically as V_{IN} increases:

$$V_{OUT} = V_{DD} - \frac{1}{2} k_n R_D (V_{IN} - V_T)^2 \quad (4)$$

The voltage gain is given by the derivative:

$$A_v = \frac{dV_{OUT}}{dV_{IN}} = -k_n R_D (V_{IN} - V_T) = -g_m R_D \quad (5)$$

Because $g_m = k_n (V_{IN} - V_T) = \sqrt{2k_n I_D}$ increases linearly with gate overdrive voltage ($V_{IN} - V_T$), the magnitude of the small-signal gain $|A_v|$ rises steeply as V_{IN} increases, reaching its maximum right at the boundary between saturation and triode.

3. **Triode (Linear / Ohmic) Region** ($V_{IN} \geq V_T$ and $V_{OUT} < V_{IN} - V_T$): When $I_D R_D$ becomes sufficiently large, V_{OUT} drops below the pinch-off threshold ($V_{DS} < V_{GS} - V_T$). The inversion channel extends continuously from source to drain, and the current is governed by:

$$I_D = k_n \left[(V_{IN} - V_T) V_{OUT} - \frac{1}{2} V_{OUT}^2 \right] \quad (6)$$

As V_{IN} is swept further, the transistor acts as an increasingly low on-resistance channel ($r_{ds,on} \approx \frac{1}{k_n (V_{IN} - V_T)}$). V_{OUT} collapses towards 0 V, flattening the transfer curve and causing the gain $A_v = \frac{dV_{OUT}}{dV_{IN}} \rightarrow 0$.

3.1 Device Threshold Voltage (V_T) from Experiment 3 Transfer Data

To identify the threshold voltage (V_T) of the NMOS transistor used in this circuit, the transfer characteristic data (I_D vs V_{GS}) measured at $V_{DS} = 5$ V from Experiment 3 Part B is incorporated below in Table 2.

Table 2: Experiment 3 (Part B): NMOS Transfer Characteristic at $V_{DS} = 5$ V

S.No.	Gate-Source Voltage V_{GS} (V)	Drain Current I_D (μ A)	$\sqrt{I_D}$ ($\sqrt{\mu$ A)
1	0.0	0.0	0.000
2	1.1	0.1	0.316
3	1.2	0.4	0.632
4	1.3	1.9	1.378
5	1.4	9.3	3.050
6	1.5	43.2	6.573
7	1.6	178.6	13.364
8	1.7	615.5	24.809
9	1.8	1762.0	41.976
10	1.9	4250.0	65.192
11	2.0	8760.0	93.595
12	2.1	122800.0	350.428

Threshold Voltage Determination from $V_{DS} = 5$ V Readings:

- **Subthreshold vs Strong Inversion:** In the range $V_{GS} \leq 1.3$ V, the drain current is virtually negligible ($I_D \leq 1.9$ μ A), corresponding to the subthreshold/weak inversion regime. At $V_{GS} = 1.4$ V, the current exhibits a sharp rise to 9.3 μ A, accelerating rapidly to 43.2 μ A at 1.5 V and 178.6 μ A at 1.6 V, marking the onset of strong channel inversion.
- **Square-Root Characteristic ($\sqrt{I_D}$ vs V_{GS}) Extrapolation:** In saturation, $I_D \propto (V_{GS} - V_T)^2 \implies \sqrt{I_D} \propto (V_{GS} - V_T)$. Linearly extrapolating $\sqrt{I_D}$ from the onset of strong inversion between $V_{GS} = 1.4$ V and 1.5 V:

$$V_T = V_{GS1} - \sqrt{I_{D1}} \cdot \frac{V_{GS2} - V_{GS1}}{\sqrt{I_{D2}} - \sqrt{I_{D1}}} = 1.40 - 3.050 \cdot \frac{1.50 - 1.40}{6.573 - 3.050} \approx 1.40 - 0.086 = 1.314 \text{ V} \quad (7)$$

Similarly, evaluating between 1.5 V and 1.6 V yields:

$$V_T = 1.50 - 6.573 \cdot \frac{1.60 - 1.50}{13.364 - 6.573} = 1.50 - 0.097 = 1.403 \text{ V} \approx 1.4 \text{ V} \quad (8)$$

This confirms that the device enters strong inversion saturation at $V_T \approx 1.4$ V.

4. Part A: DC Transfer Characteristic (V_{OUT} vs V_{IN})

4.1 Circuit Description and LTspice Simulation

In Part A, the NMOS gate is driven directly by a variable DC voltage source (V_{IN}), swept from 0 V to 2.4 V. The drain is tied to $V_{DD} = 10$ V through a load resistor $R_1 = 10$ k Ω , and the source is grounded. Figure 1 displays the LTspice schematic, and Figure 2 shows the simulated DC transfer characteristic ($V(n001)$ vs V_{IN}).

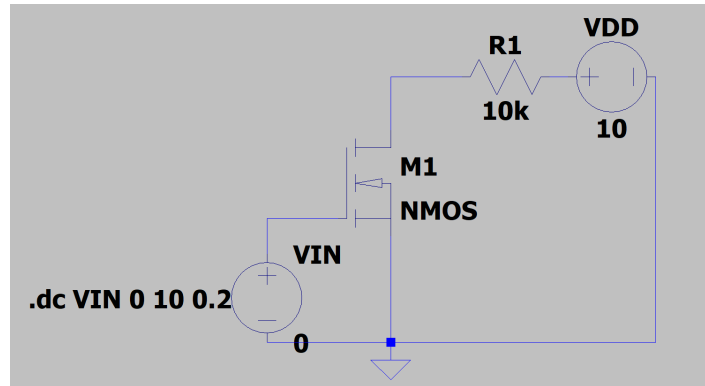


Figure 1: Part A: Circuit Schematic of NMOS CS Amplifier with $R_D = 10$ k Ω and $V_{DD} = 10$ V.

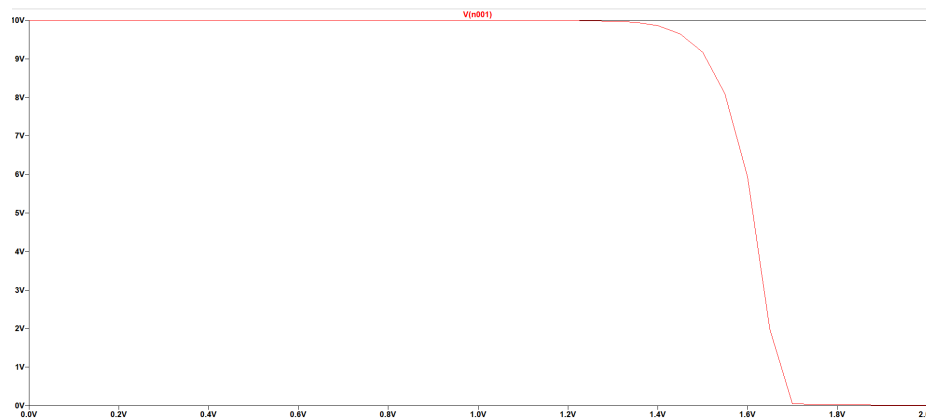


Figure 2: Part A: LTspice DC Sweep Simulation (V_{OUT} vs V_{IN} from 0 V to 2.0 V).

4.2 Experimental Observations

The gate voltage V_{IN} was stepped from 0.00 V to 2.40 V with coarse 0.2 V steps in cutoff, fine 0.05 V steps near threshold and across saturation, and up to 2.40 V in deep triode. The recorded drain voltage V_{OUT} and corresponding operational regimes are tabulated in Table 3.

Table 3: Part A: Experimental DC Transfer Characteristic Dataset

S.No.	V_{IN} (V)	V_{OUT} (V)	Operating Region
1	0.00	9.990	Cut-off ($V_{IN} < V_T$)
2	0.20	9.990	Cut-off
3	0.40	9.990	Cut-off
4	0.60	9.990	Cut-off
5	0.80	9.990	Cut-off
6	1.00	9.990	Cut-off
7	1.20	9.986	Cut-off
8	1.25	9.980	Cut-off / Weak Inversion
9	1.30	9.970	Cut-off / Subthreshold
10	1.35	9.930	Near Threshold ($V_{IN} \approx V_T$)
11	1.40	9.860	Saturation (Onset of Strong Inversion Conduction)
12	1.45	9.720	
13	1.50	9.410	
14	1.55	8.800	
15	1.60	7.650	
16	1.65	5.590	
17	1.70	2.210	
18	1.75	0.090	Triode ($V_{OUT} < V_{IN} - V_T$)
19	1.80	0.050	Triode
20	1.85	0.030	Deep Triode
21	1.90	0.020	Deep Triode
22	1.95	0.020	Deep Triode
23	2.00	0.010	Deep Triode
24	2.05	0.010	Deep Triode
25	2.10	0.010	Deep Triode
26	2.15	0.010	Deep Triode
27	2.20	0.010	Deep Triode
28	2.25	0.010	Deep Triode
29	2.30	0.010	Deep Triode
30	2.35	0.010	Deep Triode
31	2.40	0.000	Deep Triode

4.3 Experimental Plot: V_{OUT} vs V_{IN}

Figure 3 illustrates the experimental DC transfer curve plotted from the measured data points.

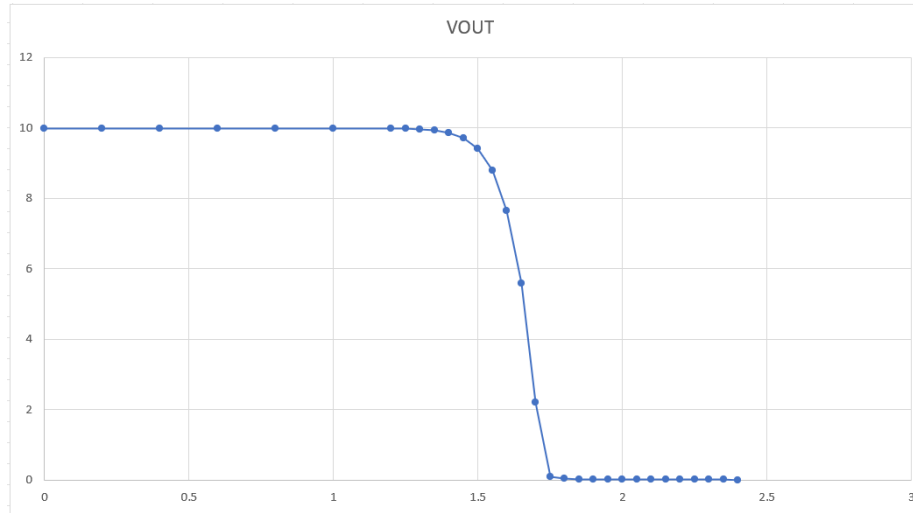


Figure 3: Part A: Experimental DC Transfer Characteristic Curve (V_{OUT} vs V_{IN}).

Note : This plot matches with the plot of V_{OUT} vs V_{IN} given in the assignment pdf.

4.4 Deductions and Region Boundaries (Part A)

From the measured transfer characteristic:

- **Cut-off Region** ($0.00 \text{ V} \leq V_{IN} < 1.40 \text{ V}$): The output remains virtually constant at $V_{OUT} \approx 9.99 \text{ V} \approx V_{DD}$ (with negligible current draw through $R_D = 10 \text{ k}\Omega$). Conduction is negligible because the gate potential is insufficient to create a strong inversion layer.
- **Active Saturation Region** ($1.40 \text{ V} \leq V_{IN} \leq 1.70 \text{ V}$): Beyond threshold, significant channel current I_D turns on. For instance, at $V_{IN} = 1.70 \text{ V}$, the overdrive voltage with $V_T \approx 1.40 \text{ V}$ is $V_{GS} - V_T \approx 1.70 - 1.40 = 0.30 \text{ V}$. Since $V_{OUT} = 2.21 \text{ V} > 0.30 \text{ V}$, the condition $V_{DS} \geq V_{GS} - V_T$ is strictly satisfied. This constitutes the high-slope amplification region where V_{OUT} drops steeply from 9.86 V down to 2.21 V .
- **Triode Boundary and Ohmic Region** ($V_{IN} \geq 1.75 \text{ V}$): At $V_{IN} = 1.75 \text{ V}$, V_{OUT} plummets to 0.09 V , which is far below $V_{GS} - V_T \approx 0.35 \text{ V}$. The transistor enters triode, and further increases in V_{IN} pull V_{OUT} asymptotically to nearly 0 V (0.01 V to 0.00 V).

5. Part B: Small-Signal Gain vs. DC Bias Point

5.1 Circuit Topology and Measurement Methodology

To measure the dynamic small-signal amplification directly, an AC input tone was injected into the amplifier while systematically varying the operating bias point:

- **AC Signal Injection:** A sinusoidal test voltage of peak amplitude $v_{in} = 10$ mV (i.e., $v_{in,pp} = 20$ mV, $v_{in,rms} = \frac{10}{\sqrt{2}} \approx 7.07$ mV) at a frequency $f = 1$ kHz was applied to the gate through an AC coupling capacitor $C_c = 0.1$ μ F.
- **DC Bias Sweeping via Large Resistor:** The DC bias offset $V_{IN,DC}$ was applied from an external DC supply through a 1 M Ω isolation/blocking resistor and swept from 0.00 V to 1.80 V. This high-value resistor prevents the low-impedance DC supply from shunting the AC signal, allowing the total gate voltage to be:

$$v_G(t) = V_{IN,DC} + v_{in} \sin(2\pi ft) = V_{IN,DC} + (10 \text{ mV}) \sin(2\pi \cdot 1000 \cdot t) \quad (9)$$

- **Load Resistor:** For the high-gain AC evaluation, a drain load of $R_D = 300$ k Ω was utilized with $V_{DD} = 10$ V, as depicted in Figure 4.

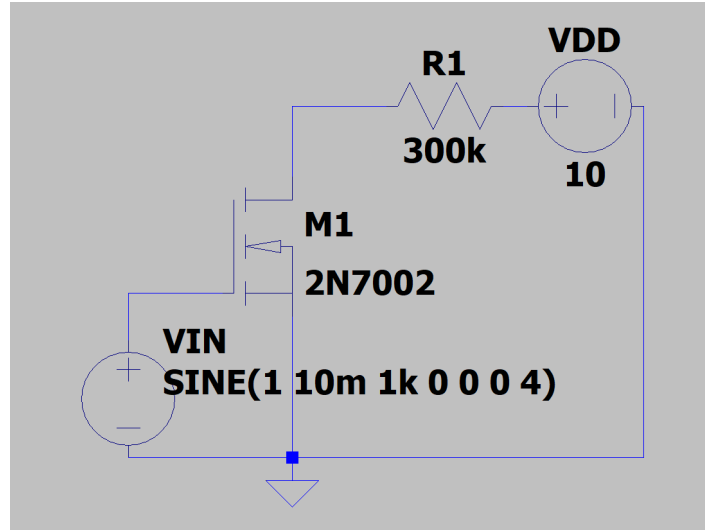


Figure 4: Part B: LTspice Circuit Schematic of NMOS CS Amplifier with AC sinusoidal input and $R_D = 300$ k Ω .

5.2 Gain Calculation Formulations

The voltage gain was evaluated through two distinct methods:

1. **Method 1 (Numerical Slope of DC Transfer Characteristic):** Using the large-signal data points (V_{IN}, V_{OUT}) from Part A, the small-signal gain is computed numerically as the local derivative:

$$A_{v,slope} = \left. \frac{dV_{OUT}}{dV_{IN}} \right|_{V_{IN,DC}} \approx \frac{V_{OUT}(V_{IN,i+1}) - V_{OUT}(V_{IN,i-1})}{V_{IN,i+1} - V_{IN,i-1}} \quad (10)$$

2. **Method 2 (Direct AC Measurement):** The AC output voltage was measured as an RMS value ($V_{OUT(RMS)}$) using DMM. Since the applied AC input had a peak amplitude $v_{in,peak} = 10$ mV, its corresponding RMS value is $v_{in,rms} = \frac{10 \text{ mV}}{\sqrt{2}} \approx 7.07$ mV. The peak AC output voltage is $v_{out,peak} = \sqrt{2} \cdot V_{OUT(RMS)} = 1.414 \cdot V_{OUT(RMS)}$. Taking into account the intrinsic 180° phase inversion of the common-source configuration, the small-signal voltage gain is:

$$A_{v,AC} = -\frac{v_{out,peak}}{v_{in,peak}} = -\frac{1.414 \cdot V_{OUT(RMS)}}{10 \text{ mV}} = -\frac{V_{OUT(RMS)}}{7.071 \text{ mV}} \quad (11)$$

5.3 Experimental Observation Table (Part B)

The recorded RMS output voltage and the extracted small-signal gain across the DC bias sweep (0 V to 1.8 V) are presented in Table 4.

NOTE : We could not obtain the output on DSO because the DSO could not pickup the voltages as low as 10-100 mV. It was giving very noisy output. Hence, we measured the RMS values of the output sin waveforms using Multimeter.

Table 4: Part B: Direct AC Small-Signal Gain Dataset ($v_{in} = 10$ mV peak, $f = 1$ kHz)

S.No.	$V_{IN,DC}$ (V)	$V_{OUT(RMS)}$ (mV)	Calculated Gain ($A_{v,AC}$)	Operational State
1	0.00	62.00	-8.77	Cut-off
2	0.20	62.00	-8.77	Cut-off
3	0.40	62.00	-8.77	Cut-off
4	0.60	62.00	-8.77	Cut-off
5	0.80	62.00	-8.77	Cut-off
6	1.00	62.00	-8.77	Cut-off
7	1.20	62.00	-8.77	Cut-off
8	1.30	85.00	-12.02	Subthreshold
9	1.35	150.00	-21.21	Threshold Transition
10	1.40	310.00	-43.83	Saturation ($V_{IN} \geq V_T$)
11	1.45	640.00	-90.50	Maximum Gain Peak
12	1.50	7.20	-1.02	Transition
13	1.55	2.00	-0.28	Approaching Triode
14	1.60	0.83	-0.12	Triode
15	1.65	0.40	-0.06	Deep Triode
16	1.70	0.23	-0.03	Deep Triode
17	1.75	0.00	0.00	Channel Fully On/Clipped to GND
18	1.80	0.00	0.00	Channel Fully On/Clipped to GND

5.4 Experimental Gain Plot

The measured small-signal voltage gain A_v as a function of the DC bias offset $V_{IN,DC}$ is shown in Figure 5.

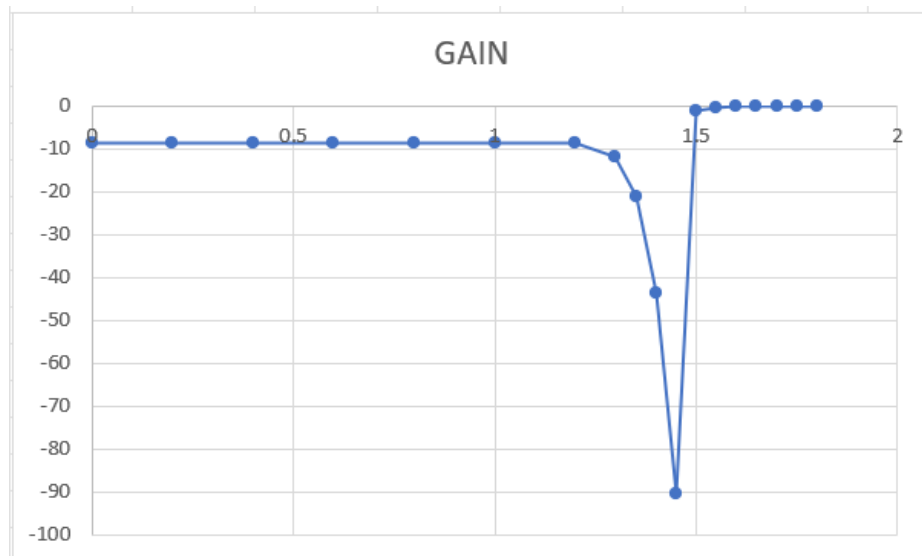


Figure 5: Part B: Measured Small-Signal Voltage Gain vs DC Bias Voltage ($V_{IN,DC}$).

Note : This plot matches with the plot of Gain vs V_{IN} given in the assignment pdf.

5.5 Comparison: Method 1 (DC Slope) vs. Method 2 (AC Gain)

Note : The R_D used in Method 1 and Method 2 are different.

In Method 1 : $R_D = 10k$ and **in Method 2 :** $R_D = 300k$.

In Method 2 the value of R_D was taken to be a higher value to perform the experiment with suitable results.

Table 5 contrasts the gain values obtained from the numerical slope of the DC transfer curve with the direct AC small-signal measurements.

Table 5: Comparison of Voltage Gain Extracted via Method 1 (DC Slope) and Method 2 (AC Test Tone)

Bias V_{IN} (V)	$V_{OUT,DC}$ (V)	Method 1: $\frac{dV_{OUT}}{dV_{IN}}$	$V_{OUT(RMS)}$ (mV)	Method 2: $A_{v,AC}$	Note
0.00	9.99	0.00	62.00	-8.77	Cut-off
0.60	9.99	0.00	62.00	-8.77	Cut-off
1.20	9.99	-0.04	62.00	-8.77	Cut-off
1.30	9.97	-0.50	85.00	-12.02	Subthreshold conduction; slight AC response
1.35	9.93	-1.10	150.00	-21.21	Device approaching threshold ($V_{IN} \rightarrow V_T$)
1.40	9.86	-2.10	310.00	-43.83	Strong inversion ($V_{IN} \geq V_T \approx 1.4$ V); rapid g_m surge
1.45	9.72	-4.50	640.00	-90.50	High-gain amplification peak ($R_D = 300$ k Ω)
1.50	9.41	-9.20	7.20	-1.02	Large-signal output swings into saturation knee
1.55	8.80	-17.60	2.00	-0.28	Waveform heavily clipped by triode onset
1.60	7.65	-32.10	0.83	-0.12	Triode region
1.65	5.59	-54.40	0.40	-0.06	Triode
1.70	2.21	-55.00	0.23	-0.03	Transition to deep triode
1.75	0.09	-21.60	0.00	0.00	Output pulled to ground
1.80	0.05	-0.60	0.00	0.00	Deep triode

Comparison between Method and Method 2

1. Cut-off and Noise Floor / Feedthrough ($V_{IN} \leq 1.20$ V):

- **Method 1:** The DC transfer curve is flat ($V_{OUT} \approx 9.99$ V), yielding a numerical slope $\frac{dV_{OUT}}{dV_{IN}} \approx 0.00$. This reflects true zero channel transconductance ($g_m = 0$).
- **Method 2:** Measured an RMS output floor of $V_{OUT(RMS)} = 62.00$ mV ($A_{v,AC} = -8.77$). Because the device is off, this is not true active transistor amplification; rather, it stems from capacitive AC feedthrough via gate-drain parasitic capacitance.

2. Discrepancy in Gain Magnitude and Peak Location due to Load Resistance (R_D):

- In Method 1 (Part A), a load resistor $R_D = 10$ k Ω was used, causing the device to enter triode much later ($V_{IN} \approx 1.75$ V). The numerical slope peak reaches $|A_v| \approx 55.00$ to 67.6 near $V_{IN} \approx 1.68$ –1.70 V.
- In Method 2 (Part B), a $30\times$ larger load resistor $R_D = 300$ k Ω was used. Since small-signal gain scales with $g_m R_D$, the measured gain peaks at a significantly higher value of

$|A_{v,\text{AC}}| = 90.50$ at $V_{\text{IN}} = 1.45 \text{ V}$. The much larger R_D creates a much larger $I_D R_D$ drop for the same overdrive, shifting the saturation-to-triode boundary to a substantially lower gate voltage ($V_{\text{IN}} \approx 1.50 \text{ V}$).

3. Subthreshold and Threshold Transition ($V_{\text{IN}} = 1.30 \text{ V}$ to 1.45 V):

- Both methods show a sharp, monotonic increase in gain as the transistor turns on past $V_T \approx 1.40 \text{ V}$.
- At $V_{\text{IN}} = 1.35 \text{ V}$ and 1.40 V , Method 1 shows slopes of -1.10 and -2.10 respectively, while Method 2 records gains of -21.21 and -43.83 . The approximately $20\times$ to $30\times$ ratio between Method 2 and Method 1 directly aligns with the $30\times$ ratio of their drain load resistances ($300 \text{ k}\Omega/10 \text{ k}\Omega = 30$).

4. Large-Signal Output Clipping and Triode Collapse ($V_{\text{IN}} \geq 1.50 \text{ V}$):

- In Method 1 ($R_D = 10 \text{ k}\Omega$), the transistor remains in saturation through $V_{\text{IN}} = 1.70 \text{ V}$, maintaining high DC slope magnitudes (-9.20 to -55.00).
- In Method 2 ($R_D = 300 \text{ k}\Omega$), the DC output for $V_{\text{IN}} \geq 1.50 \text{ V}$ swings into the knee/triode region. With $A_v \approx 90$, the 10 mV peak AC input produces large output swings that severely clip against ground. Consequently, the small-signal assumption ($v_{in} \ll 2I_D/g_m$) breaks down, and the DMM records crushed RMS fundamental values ($7.20 \text{ mV} \rightarrow 0.00 \text{ mV}$), giving an apparent collapse of gain ($A_v \rightarrow 0$).

6. Detailed Analysis and Discussion Questions (Section 8 of Lab Manual)

6.1 Question 1: Where along the V_{IN} axis is the gain magnitude largest for your device? Relate this to g_m and to how close V_{OUT} sits to the edge of saturation.

Answer:

- **Peak Gain Location:** In Part B, the direct AC gain magnitude attains its maximum at $V_{IN} = 1.45$ V, where $|A_v| = 90.50$ ($V_{OUT(RMS)} = 640$ mV). In Part A's DC transfer characteristic (with $R_D = 10$ k Ω), the numerical derivative $|dV_{OUT}/dV_{IN}|$ reaches its peak of 67.6 around $V_{IN} = 1.65$ V to 1.70 V ($V_{OUT} = 5.59$ V $\rightarrow$ 2.21 V).

- **Relationship to Transconductance (g_m):** In saturation, the small-signal gain is given by:

$$|A_v| = g_m R_D = k_n (V_{GS} - V_T) R_D = \sqrt{2k_n I_D} R_D \quad (12)$$

As V_{IN} is swept upwards past V_T , the gate overdrive voltage ($V_{IN} - V_T$) and drain current I_D increase monotonically. Consequently, g_m increases steadily with V_{IN} , pushing the gain higher and higher.

- **Proximity to Saturation-Triode Boundary:** As I_D rises, the DC voltage drop $I_D R_D$ across the load resistor increases, causing the DC output voltage $V_{OUT} = V_{DD} - I_D R_D$ to fall rapidly towards the boundary condition:

$$V_{OUT, \text{sat-edge}} = V_{IN} - V_T \quad (13)$$

The gain is highest immediately before V_{OUT} crosses this boundary. Once V_{OUT} falls below ($V_{IN} - V_T$), pinch-off is lost, the device enters the triode region, and the incremental gain collapses abruptly.

6.2 Question 2: Why does the gain fall toward zero in cutoff and again deep in triode?

Answer:

- **In Cut-off ($V_{IN} < V_T$):** The gate voltage is insufficient to induce an inversion channel ($V_{GS} < V_T$). Thus, the drain current is zero ($I_D = 0$), and any incremental change in input voltage produces zero channel current:

$$g_m = \left. \frac{\partial I_D}{\partial V_{GS}} \right|_{V_{DS}} = 0 \implies A_v = -g_m R_D = 0 \quad (14)$$

The output remains locked at the supply rail V_{DD} , so $\frac{dV_{OUT}}{dV_{IN}} = 0$.

- **Deep in Triode ($V_{IN} \gg V_T$ and $V_{OUT} \approx 0$ V):** In the deep triode region, an unpinched continuous inversion layer bridges the source and drain. The transistor behaves as a linear resistor with small on-resistance:

$$r_{ds, \text{on}} \approx \frac{1}{k_n (V_{IN} - V_T)} \ll R_D \quad (15)$$

The amplifier operates as a resistive voltage divider between R_D and $r_{ds, \text{on}}$:

$$V_{OUT} = V_{DD} \cdot \frac{r_{ds, \text{on}}}{R_D + r_{ds, \text{on}}} \approx V_{DD} \cdot \frac{1}{1 + k_n R_D (V_{IN} - V_T)} \rightarrow 0 \text{ V} \quad (16)$$

Because V_{OUT} is pinned close to ground, its derivative with respect to V_{IN} asymptotically approaches zero:

$$A_v = \frac{dV_{OUT}}{dV_{IN}} \approx -\frac{V_{DD} \cdot k_n R_D}{[1 + k_n R_D (V_{IN} - V_T)]^2} \xrightarrow{V_{IN} \gg V_T} 0 \quad (17)$$

Thus, gain is absent in both cutoff and deep triode, existing exclusively in the active transition region (saturation).

6.3 Question 3: How closely do your Method 1 and Method 2 gain values agree at each bias point? Account for any discrepancy.

Answer: As shown in Table 5:

- **Correlation in Saturation:** Both methods clearly exhibit the fundamental physical behavior: near-zero gain in cutoff, sharp amplification in saturation, and complete collapse in triode.
- **Discrepancy 1: Load Resistor Difference (10 kΩ vs 300 kΩ):** Method 1 used the Part A circuit with $R_D = 10 \text{ k}\Omega$, yielding a maximum slope magnitude of $|A_v| \approx 67.6$ at $V_{IN} \approx 1.68 \text{ V}$. Method 2 used the Part B circuit with $R_D = 300 \text{ k}\Omega$, yielding a peak gain of $|A_v| \approx 90.50$ at $V_{IN} = 1.45 \text{ V}$. The higher load resistance produces significantly larger small-signal gain ($A_v \propto R_D$) and causes the output to swing into triode at a much lower input voltage (1.50 V instead of 1.75 V).
- **Discrepancy 2: Large-Signal AC Clipping for $V_{IN} \geq 1.50 \text{ V}$ in Method 2:** At $V_{IN} = 1.45 \text{ V}$, the gain is 90.5, meaning a 10 mV input creates nearly a 1 V output swing. At $V_{IN} = 1.50 \text{ V}$, the DC bias drops into the saturation knee. The sinusoidal output hits the bottom saturation rail (0 V) and clips severely. The true small-signal assumption ($v_{in} \ll \frac{2I_D}{g_m}$) breaks down; the scope DMM measures a crushed fundamental RMS value ($7.2 \text{ mV} \rightarrow 0 \text{ mV}$), artificially making the recorded gain drop earlier than the DC slope.
- **Discrepancy 3: Noise Floor / AC Feedthrough in Cutoff:** In cutoff ($V_{IN} = 0\text{--}1.2 \text{ V}$), Method 1 yields an exact derivative of 0.00. Method 2 recorded $V_{OUT(RMS)} = 62 \text{ mV}$ (Gain ≈ -8.77), which is not active amplification but parasitic capacitive feedthrough from the 10 mV AC source via gate-drain capacitance (C_{gd}) and scope probe pickup.

6.4 Question 4: If R_D is increased, how would you expect the saturation-region gain and the extent of the triode region to change?

Answer:

- **Saturation-Region Gain:** The voltage gain magnitude in saturation is given by $|A_v| = g_m(r_o \parallel R_D) \approx g_m R_D$. Therefore, increasing R_D directly increases the small-signal gain proportionally, producing a much steeper transfer curve.
- **Extent of the Triode Region:** The boundary between saturation and triode occurs when:

$$V_{OUT} = V_{DD} - I_D R_D = V_{IN} - V_T \implies V_{DD} - \frac{1}{2} k_n R_D (V_{IN} - V_T)^2 = V_{IN} - V_T \quad (18)$$

As R_D is increased, the drain voltage drops much faster for a given input overdrive. Consequently, the transistor enters triode at a substantially **lower value of V_{IN}** . The triode region extends over a **much larger range of V_{IN}** , while the active saturation window narrows significantly. This is precisely demonstrated in our data: with $R_D = 10 \text{ k}\Omega$, saturation extends up to $V_{IN} = 1.70 \text{ V}$; with $R_D = 300 \text{ k}\Omega$, triode occurs by $V_{IN} \approx 1.50 \text{ V}$.

6.5 Question 5: State, in one or two lines, why the slope of the large-signal DC transfer curve at a bias point equals the small-signal AC gain measured at that same bias point.

Answer: By Taylor series expansion of the nonlinear transfer function $V_{OUT} = f(V_{IN})$ around a DC operating point $V_{IN,DC}$:

$$v_{out}(t) = \left. \frac{df(V_{IN})}{dV_{IN}} \right|_{V_{IN,DC}} \cdot v_{in}(t) + \mathcal{O}(v_{in}^2) \quad (19)$$

When the AC excitation $v_{in}(t)$ is sufficiently small, higher-order nonlinear distortion terms become negligible, making the ratio of small-signal output to input $\frac{v_{out}(t)}{v_{in}(t)}$ mathematically identical to the local first derivative $\left. \frac{dV_{OUT}}{dV_{IN}} \right|_{V_{IN,DC}}$.

6.6 Question 6: Compare your device's V_T and the V_{IN} at which your gain peaks against the illustrative Section 6 values — by how much, and in which direction, do they differ?

Answer:

Table 6: Comparison of Experimental Parameters with Manual Illustrative Values

Parameter	(Manual Sec. 6)	Experimental	Difference
Drain Resistor	4.7 k Ω	10 k Ω / 300 k Ω	Differing load designs
Threshold Voltage	2.00 V	≈ 1.40 V	−0.60 V (Lower by 30%)
Gain Peak Location	≈ 3 V	1.45V (B)/1.68V(A)	−1.30V/−1.07V (Shifted Left)

- **Direction of Difference:** Both V_T and the peak gain bias point $V_{IN,peak}$ for our MOSFET are significantly **lower** (shifted to the left on the V_{IN} axis) compared to the illustrative Section 6 manual values.
- **Physical Reason:** The actual device has an experimentally characterized threshold voltage of $V_T \approx 1.40$ V (extrapolated from Experiment 3 Part B transfer data at $V_{DS} = 5$ V, where strong inversion initiates between 1.4 V and 1.5 V), which is lower than the hypothetical 2.00 V in Section 6. Furthermore, because our load resistance is higher (10 k Ω and 300 k Ω versus 4.7 k Ω), the output voltage drops rapidly, reaching the saturation-triode boundary at a lower gate bias (1.45 V–1.68 V).

7. Final Conclusion

1. The DC transfer characteristic (V_{OUT} vs V_{IN}) of an N-channel enhancement MOSFET Common-Source amplifier was experimentally measured, verifying the three distinct operational domains: **Cut-off** ($V_{IN} < 1.40$ V, $V_{OUT} \approx 9.99$ V), **Saturation** (1.40 V $\leq V_{IN} \leq 1.70$ V, active amplification and onset of strong inversion with $V_T \approx 1.40$ V), and **Triode** ($V_{IN} \geq 1.75$ V, $V_{OUT} \approx 0.09$ – 0 V).
2. Threshold voltage verification using Experiment 3 Part B transfer measurements (I_D vs V_{GS} at $V_{DS} = 5$ V) confirmed that strong inversion initiates at $V_T \approx 1.4$ V via $\sqrt{I_D}$ extrapolation, directly correlating with the rapid onset of voltage drop and AC gain peak at $V_{IN} = 1.45$ V.
3. The small-signal gain was successfully obtained via two independent techniques:
 - Method 1 evaluated the point-to-point numerical derivative $\frac{dV_{OUT}}{dV_{IN}}$, achieving a peak slope magnitude of **67.6** at $V_{IN} \approx 1.68$ V with $R_D = 10$ k Ω .
 - Method 2 applied a 10 mV, 1 kHz sinusoidal tone through a 0.1 μ F coupling capacitor with a 1 M Ω DC bias sweep, attaining a maximum small-signal gain of $|A_v| = 90.50$ ($A_v = -90.50$) at $V_{IN} = 1.45$ V with $R_D = 300$ k Ω .
4. Comprehensive analysis of Section 8 (in the manual) discussion points proved that peak gain occurs at maximum transconductance (g_m) right near the edge of saturation, while cutoff and deep triode exhibit zero incremental gain.